

AI FITTRACK API

Data Flow Diagram

Information flow between client, API, database, and AI

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DATE	30 September 2026

Context

Client/Postman -> Express API -> MongoDB for users/workouts. Protected routes verify JWT. AI requests call Google Gemini and return text.

Authentication

Registration validates fields, checks duplicate email, creates a User (password hashing runs in model hook), stores it, and returns a token. Login finds a user, compares password, and returns a token.

Workout flow

Bearer token -> auth middleware loads current user -> controller validates and scopes query -> Mongoose reads/writes MongoDB -> JSON response.

AI flow

Authenticated request -> controller validates fields -> geminiService calls SDK -> generated text is returned.

Data handling

Passwords are bcrypt-hashed; settings use environment variables. Do not publish .env values; .env is excluded from this documentation package.